Calculate $18^{13} \mod 39$ step-by-step:

1. **Square and reduce repeatedly:**
   - Compute $18^2 \mod 39$:
     $$18^2 = 324 \quad \Rightarrow \quad 324 \mod 39 = 12$$
     So, $18^2 \equiv 12 \mod 39$.
   - Compute $18^4 \mod 39$ using $(18^2)^2 \mod 39$:
     $$12^2 = 144 \quad \Rightarrow \quad 144 \mod 39 = 27$$
     So, $18^4 \equiv 27 \mod 39$.
   - Compute $18^8 \mod 39$ using $(18^4)^2 \mod 39$:
     $$27^2 = 729 \quad \Rightarrow \quad 729 \mod 39 = 27$$
     So, $18^8 \equiv 27 \mod 39$.
2. **Combine powers to reach $18^{13} \mod 39$:**
   - Break 13 into $8 + 4 + 1$:
     $$18^{13} = 18^8 \cdot 18^4 \cdot 18^1 \mod 39$$
   - Substitute the results:
     $$18^8 \equiv 27, \quad 18^4 \equiv 27, \quad 18^1 \equiv 18$$
   - Combine:
     $$27 \cdot 27 \mod 39 = 729 \mod 39 = 27$$
     Then:
     $$27 \cdot 18 \mod 39 = 486 \mod 39 = 18$$

Final Answer:

$$18^{13} \mod 39 = 18$$

This method ensures the numbers stay manageable by reducing modulo 39 at each step.

Example 1: $5^{12} \mod 13$

Step 1: Simplify the base

The base 5 is already smaller than 13. So, no reduction is needed.

Step 2: Use modular exponentiation

1. Compute $5^2 \mod 13$:

$$5^2 = 25 \quad \Rightarrow \quad 25 \mod 13 = 12$$

   So, $5^2 \equiv 12 \mod 13$.

2. Compute $5^4 \mod 13$ using $(5^2)^2 \mod 13$:

$$12^2 = 144 \quad \Rightarrow \quad 144 \mod 13 = 1$$

   So, $5^4 \equiv 1 \mod 13$.

3. Simplify $5^{12} \mod 13$ using $(5^4)^3 \mod 13$:

$$(5^4)^3 = 1^3 = 1 \quad \Rightarrow \quad 5^{12} \equiv 1 \mod 13$$

Final Answer:

$$5^{12} \mod 13 = 1$$

Example 2: $7^{10} \mod 15$

Step 1: Simplify the base

The base 7 is already smaller than 15.

Step 2: Use modular exponentiation

1. Compute $7^2 \mod 15$:

$$7^2 = 49 \quad \Rightarrow \quad 49 \mod 15 = 4$$

   So, $7^2 \equiv 4 \mod 15$.

2. Compute $7^4 \mod 15$ using $(7^2)^2 \mod 15$:

$$4^2 = 16 \quad \Rightarrow \quad 16 \mod 15 = 1$$

   So, $7^4 \equiv 1 \mod 15$.

3. Simplify $7^{10} \mod 15$ using $(7^4)^2 \cdot 7^2 \mod 15$:

$$1^2 \cdot 4 = 4 \quad \Rightarrow \quad 7^{10} \equiv 4 \mod 15$$

Final Answer:

$$7^{10} \mod 15 = 4$$

Example 3: $11^9 \mod 20$

Step 1: Simplify the base

The base 11 is already smaller than 20.

Step 2: Use modular exponentiation

1. Compute $11^2 \mod 20$:

$$11^2 = 121 \quad \Rightarrow \quad 121 \mod 20 = 1$$

   So, $11^2 \equiv 1 \mod 20$.

2. Simplify $11^9 \mod 20$ using $(11^2)^4 \cdot 11 \mod 20$:

$$1^4 \cdot 11 = 11 \quad \Rightarrow \quad 11^9 \equiv 11 \mod 20$$

Final Answer:

$$11^9 \mod 20 = 11$$